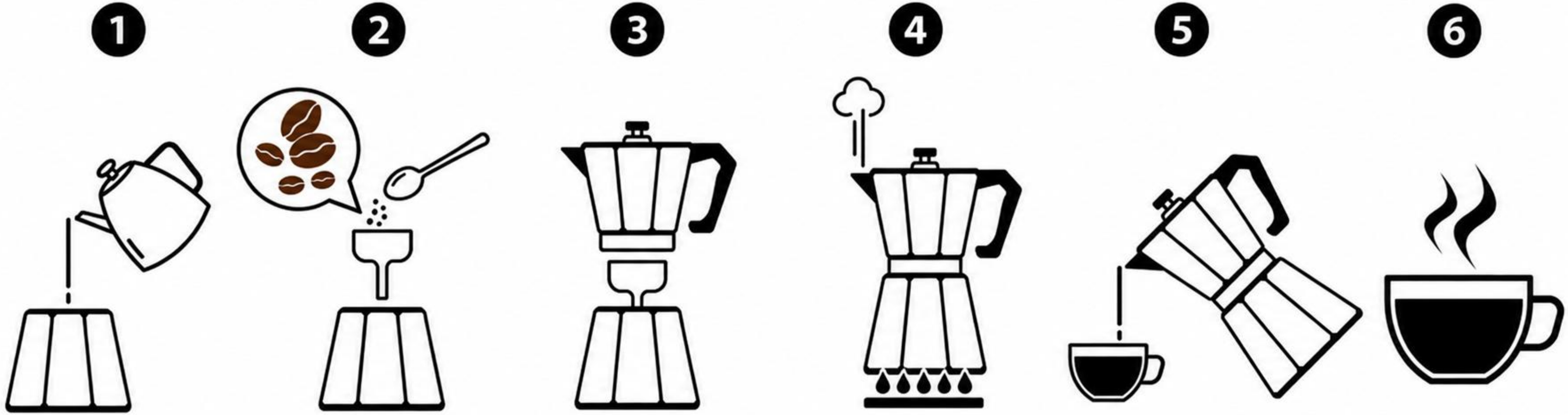
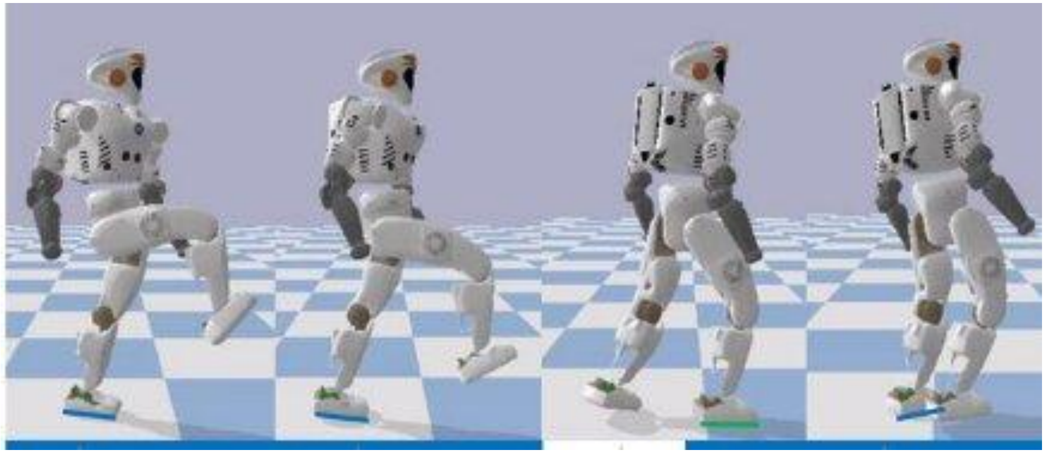
By Dian Kruger and Shane Josias

PLANNING AS AUTOREGRESSIVE SEQUENCE MODELLING: DECODING STRATEGIES



WHAT IS PLANNING?

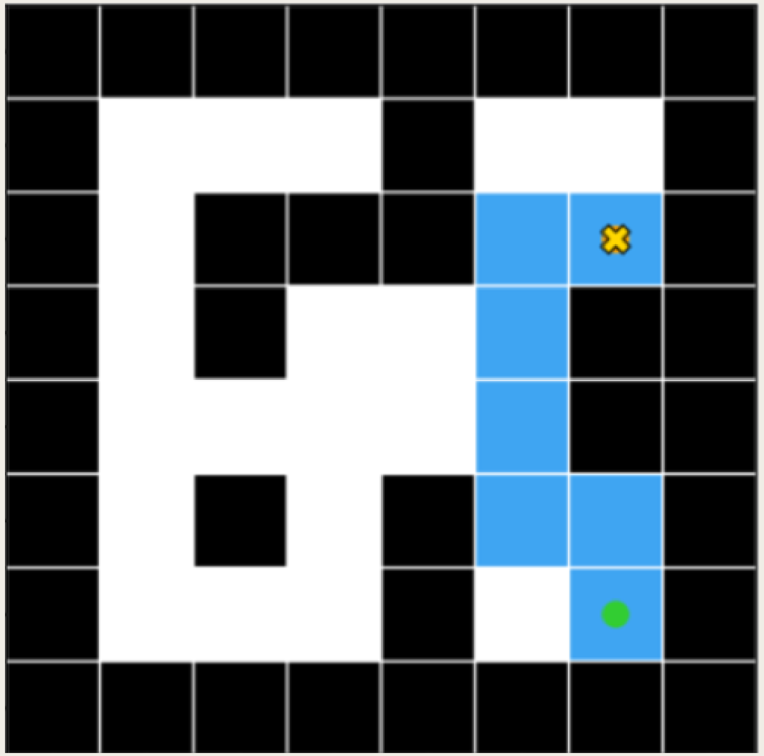
Planning is deciding which actions to take, and in what order so that an environment changes from the initial state to the goal state.



WHY LEARN PLANNING FROM DATA?

- Machines can learn to automatically solve planning problems.
- This can be done through trial and error, using techniques like reinforcement learning.
- Learning through trial and error can be risky in safety-critical or expensive settings.

THIS WORK'S PLANNING PROBLEM

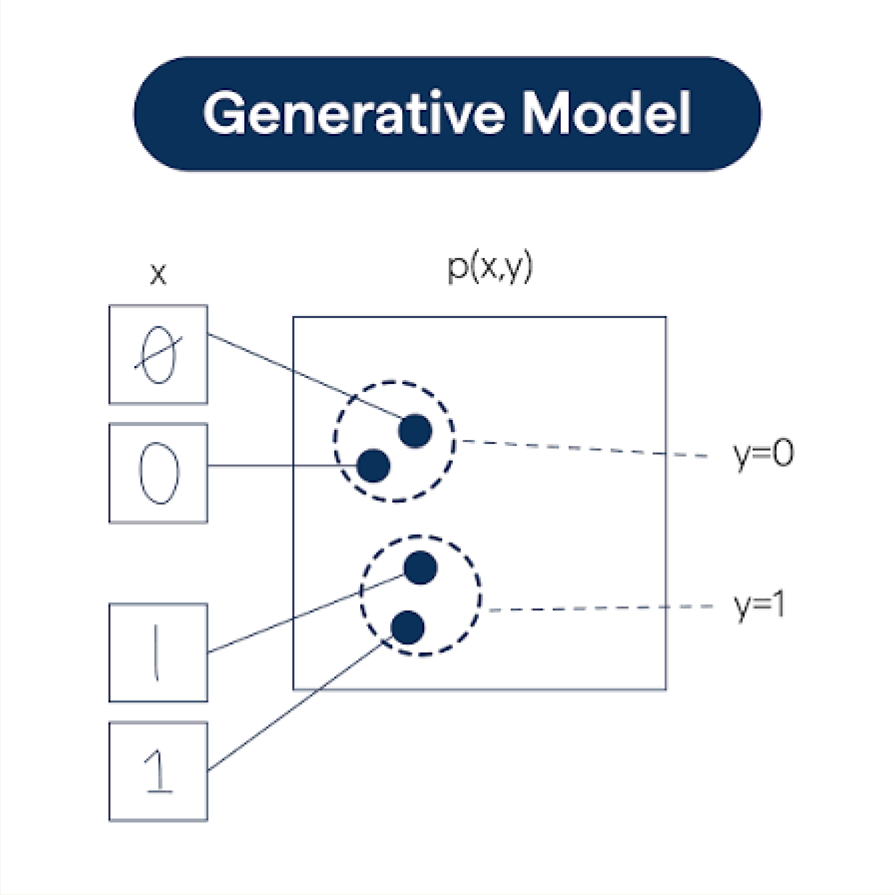


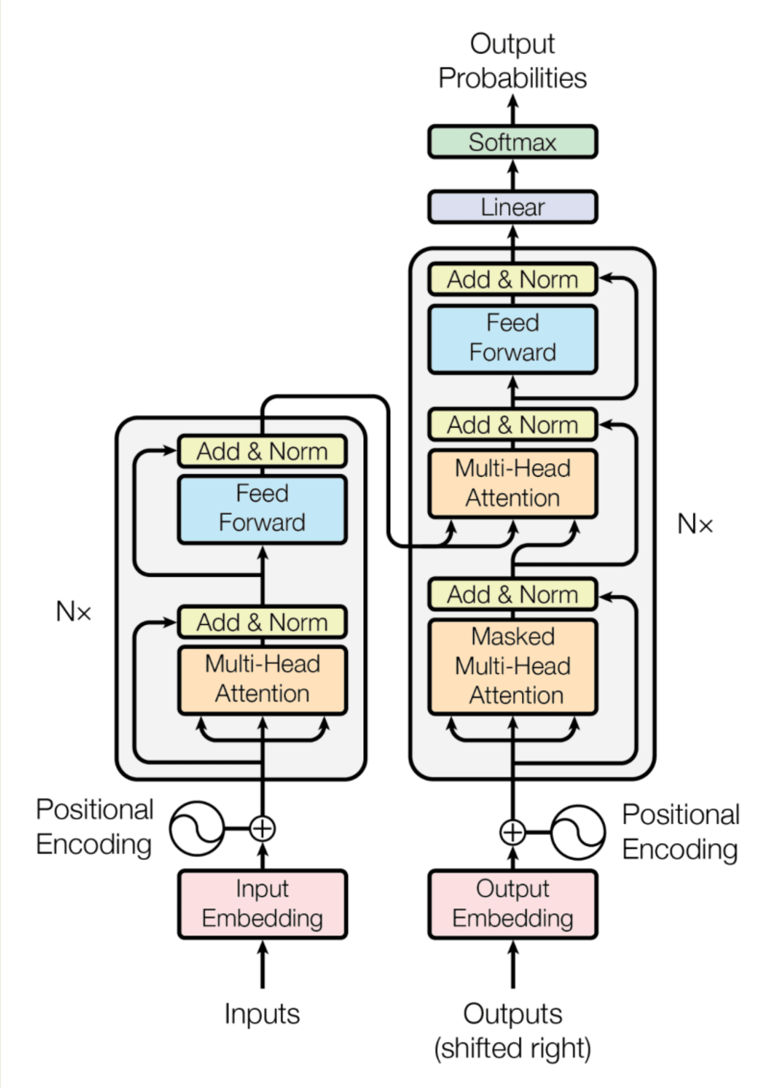
- The environment is a discrete 2D maze represented as a grid.
- A move can go only up, down, left, or right.
- A start square and a goal square are given.
- The objective is to generate a valid sequence of moves that reaches the goal without passing through walls.

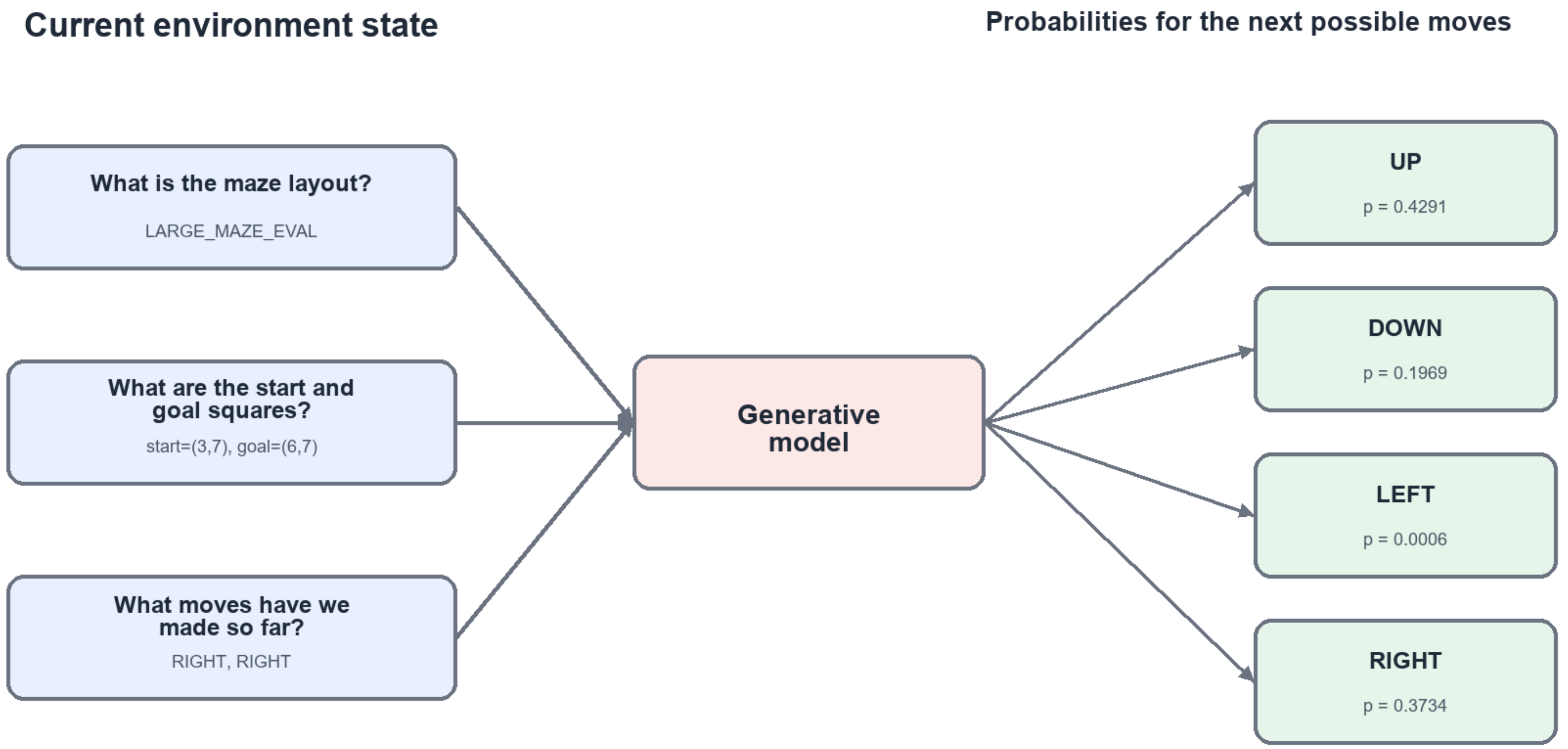
IMITATION LEARNING IN THE MAZE

Maze layout	Start -> Goal	State sequence	Action sequence
OPEN	(2,2) -> (4,4)	[(2,2),(2,3),(2,4),(3,4),(4,4)]	[UP, UP, RIGHT, RIGHT]
U_MAZE	(2,2) -> (5,3)	[(2,2),(3,2),(4,2),(5,2),(5,3)]	[RIGHT, RIGHT, RIGHT, UP]
MEDIUM_MAZE	(1,1) -> (4,2)	[(1,1),(1,2),(2,2),(2,3),(3,3),(4,3),(4,2)]	[UP, RIGHT, UP, RIGHT, RIGHT, DOWN]
MEDIUM_MAZE_EVAL	(1,1) -> (3,3)	[(1,1),(1,2),(1,3),(1,4),(2,4),(3,4),(3,3)]	[UP, UP, UP, RIGHT, RIGHT, DOWN]
LARGE_MAZE	(1,1) -> (3,3)	[(1,1),(1,2),(1,3),(2,3),(3,3)]	[UP, UP, RIGHT, RIGHT]

GENERATIVE MODELLING

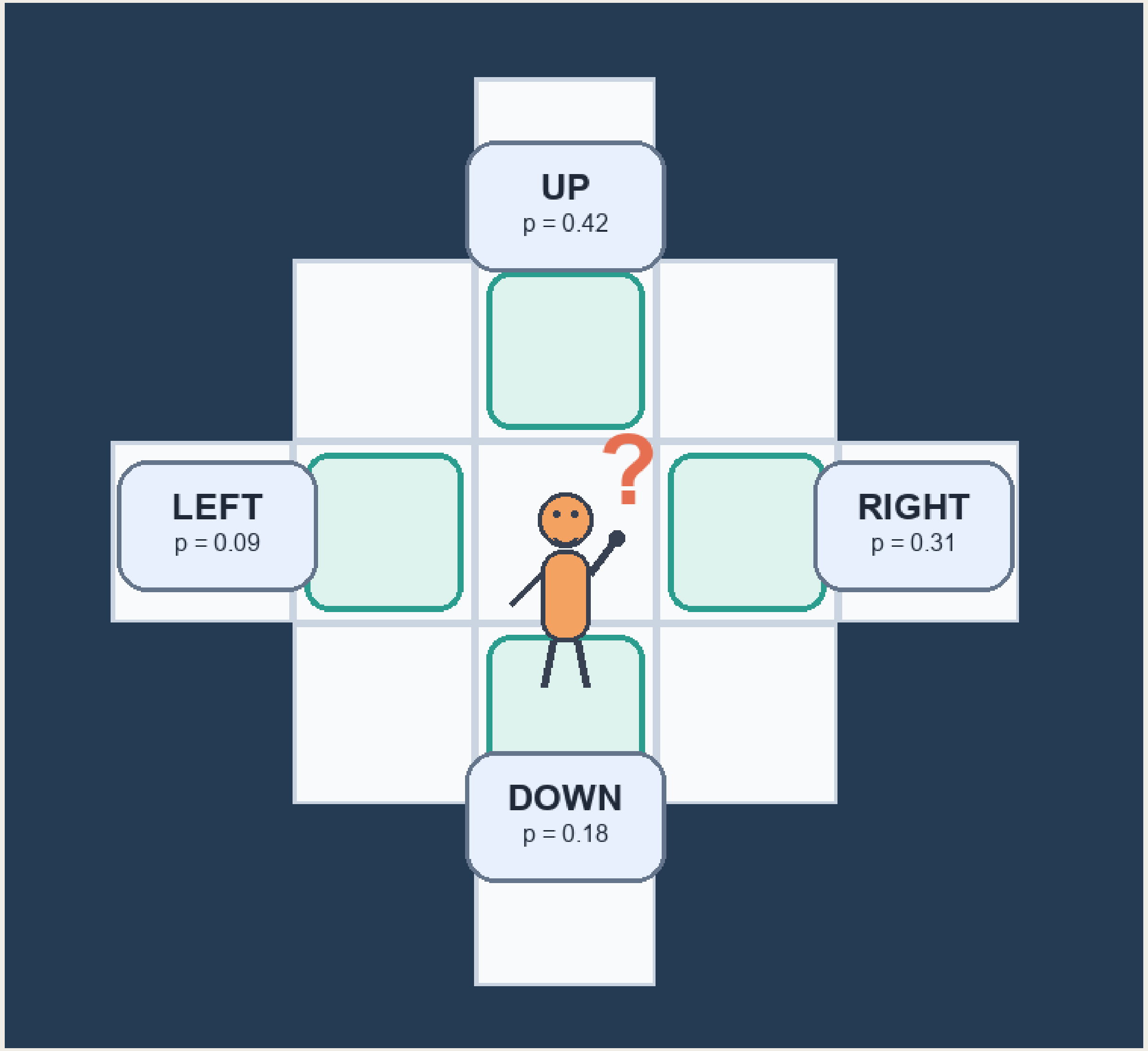




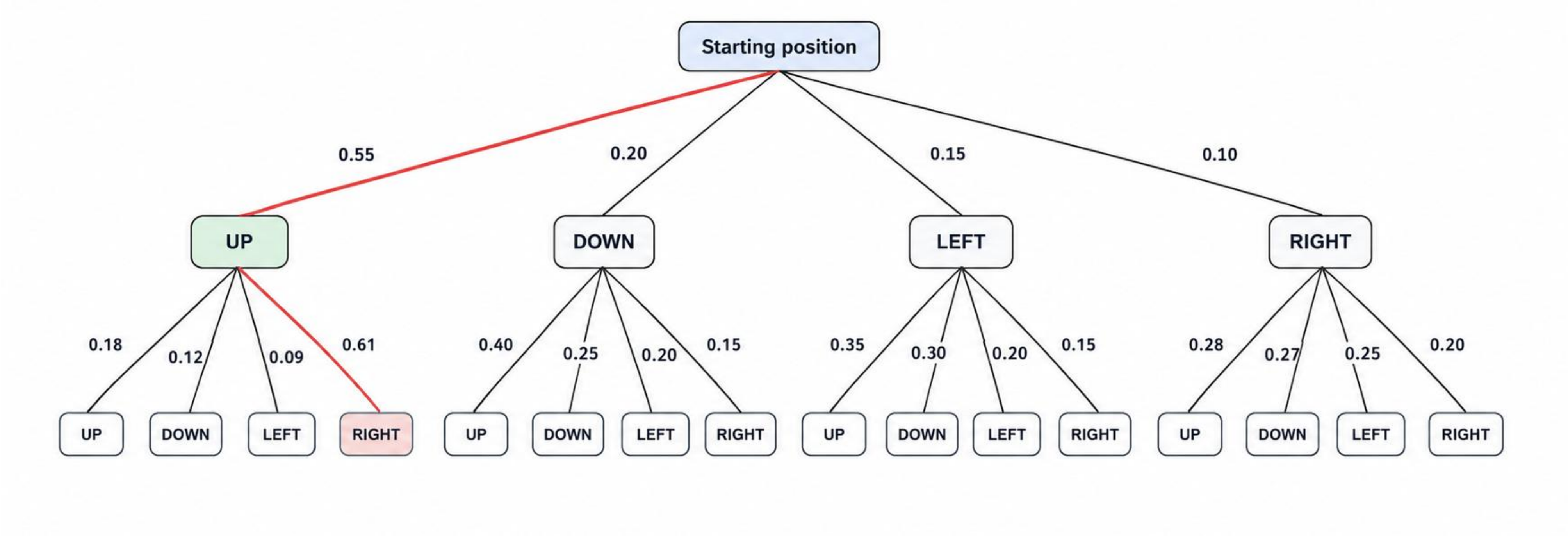


AUTOREGRESSIVE SEQUENCE MODELLING

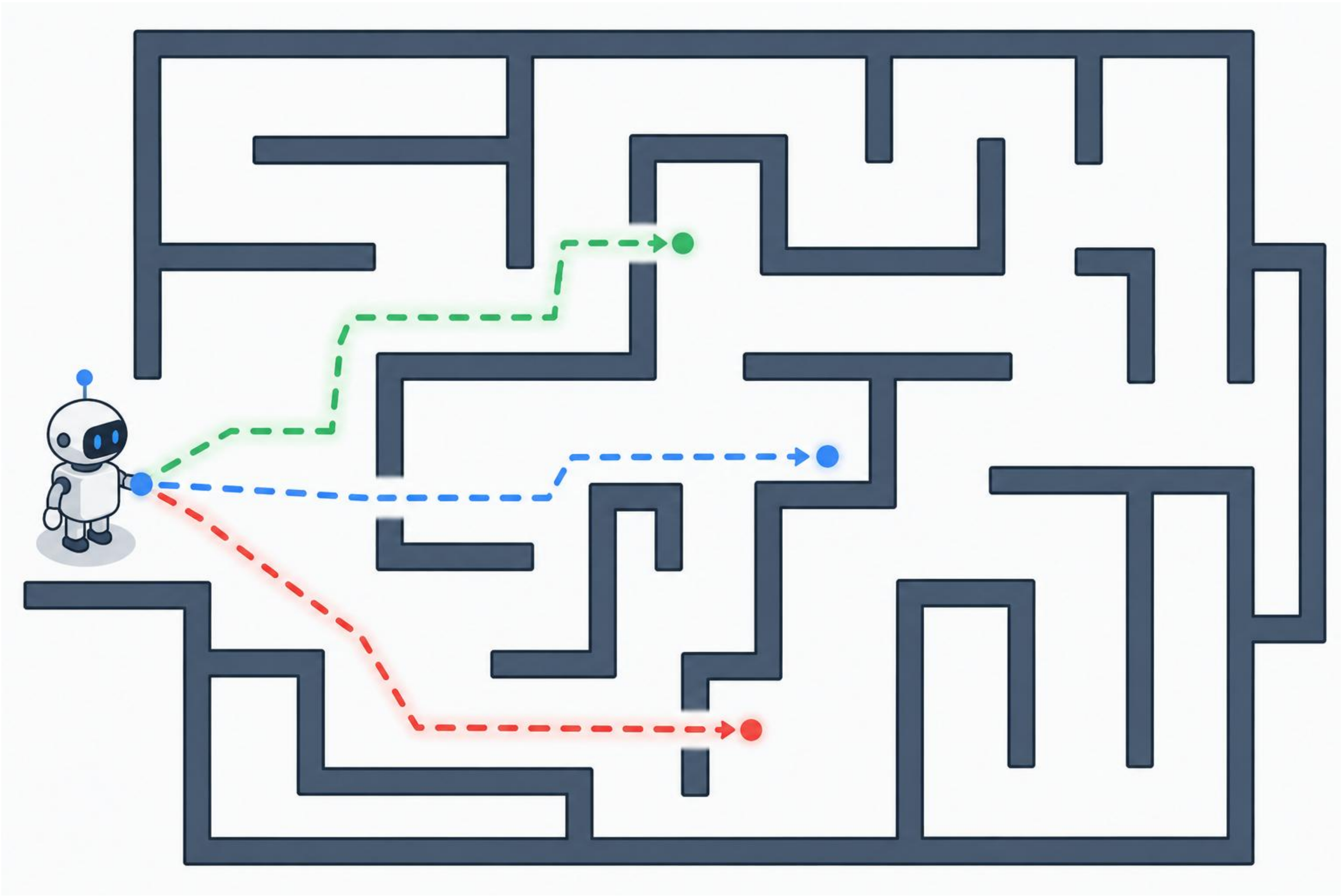
DECODING ALGORITHMS



GREEDY DECODING



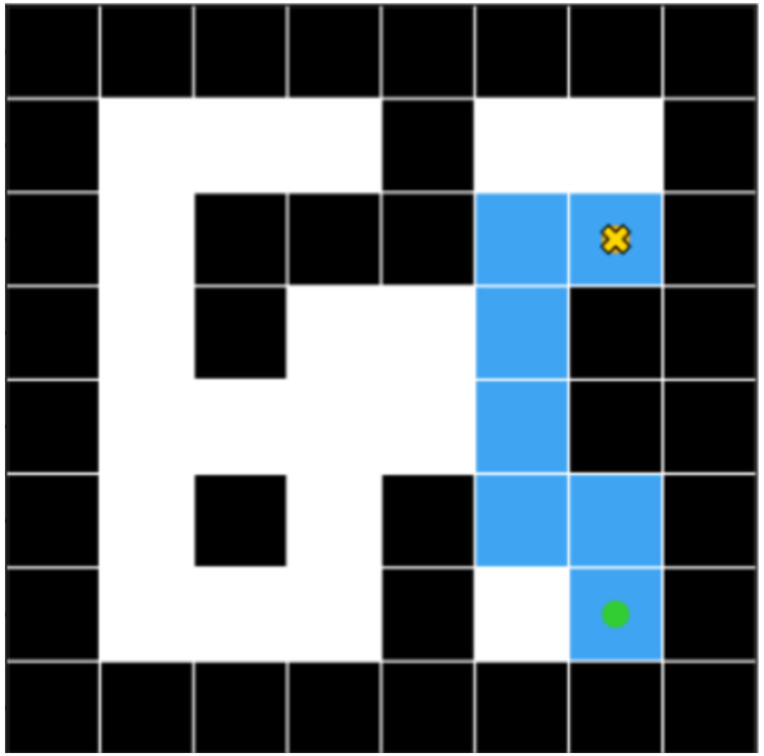
BEAM SEARCH



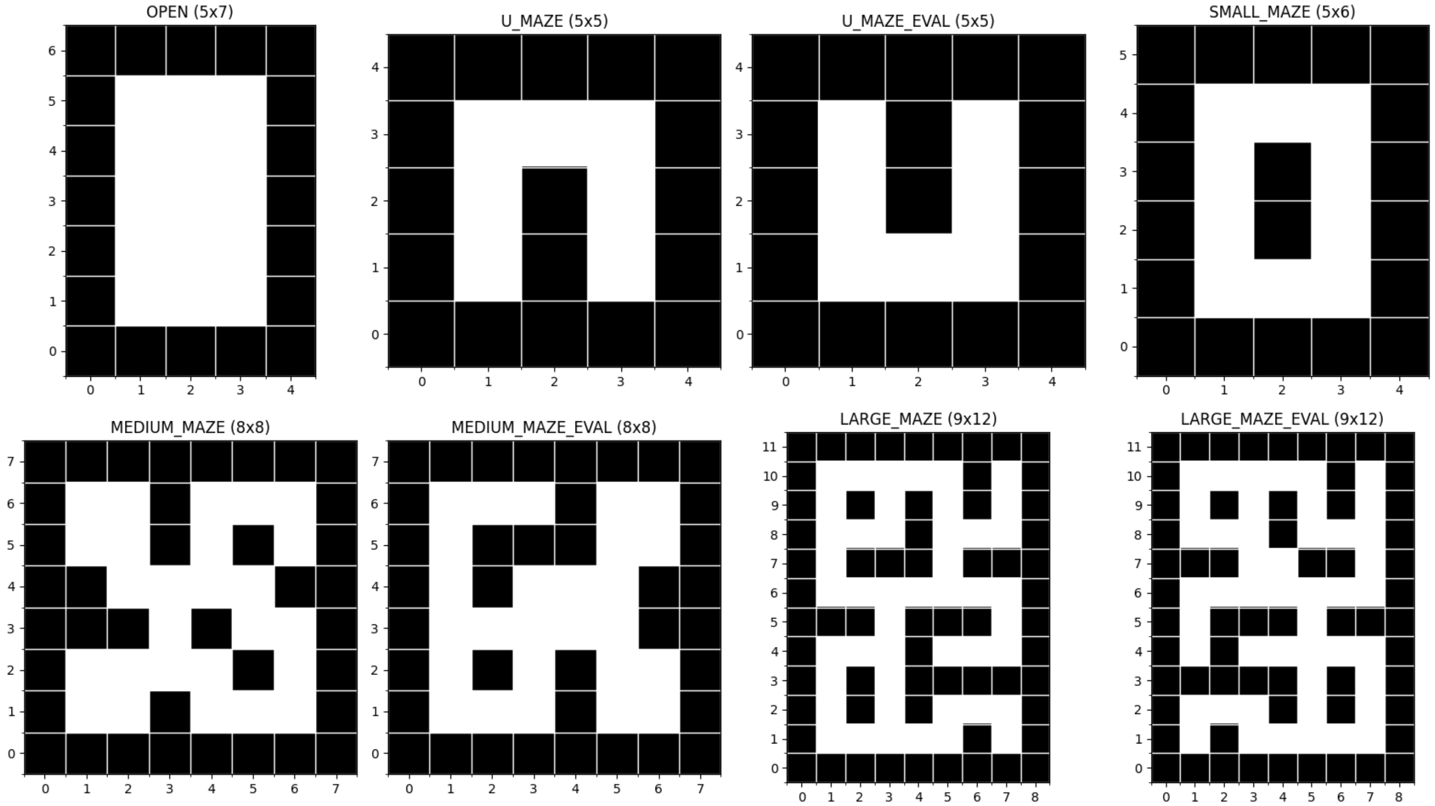
EVALUATION METRICS

- Completion Rate
- Average Steps to Goal
- Average Compute Time
- Average Time per Step

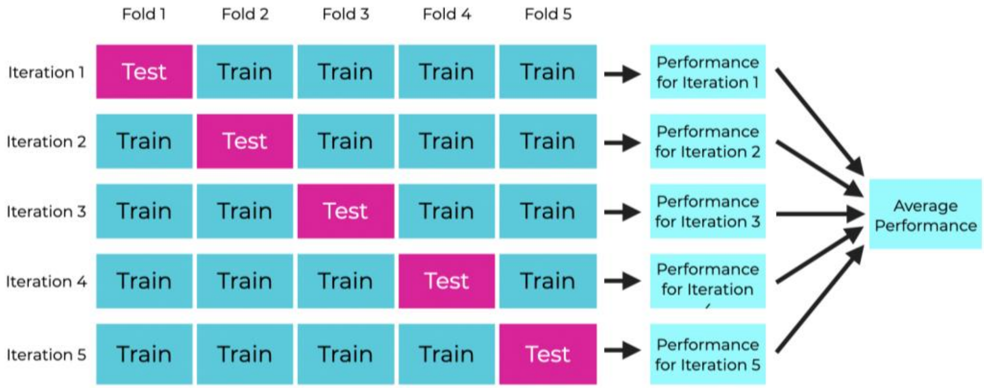
DATASET AND EVALUATION APPROACH



example of a training trajectory

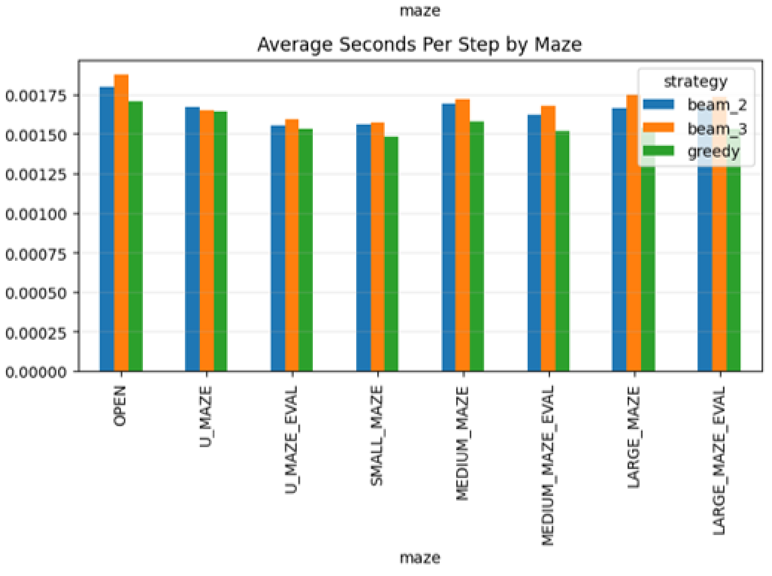


The eight maze layouts

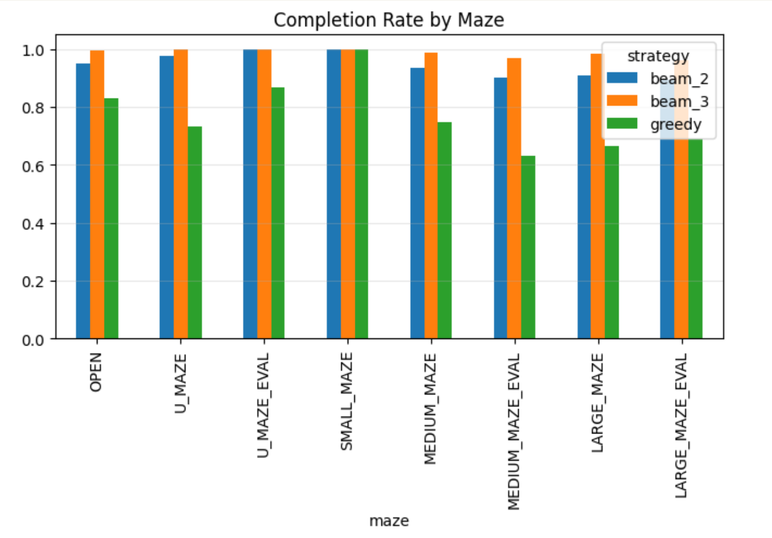


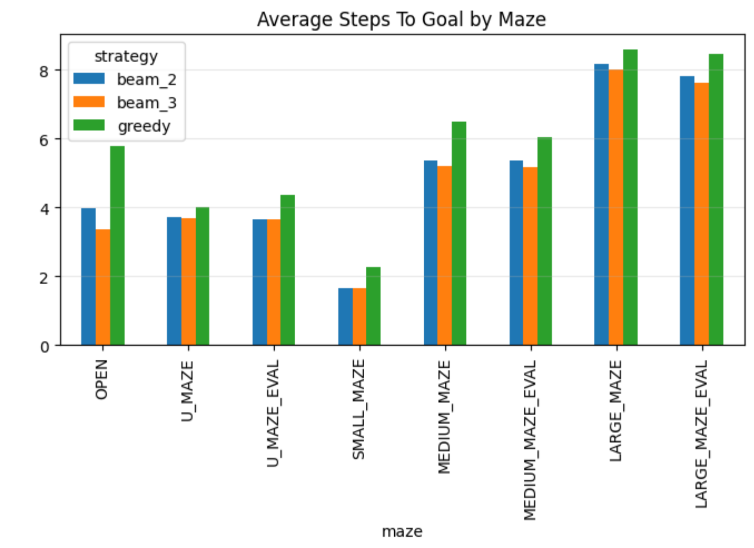
5-fold cross validation

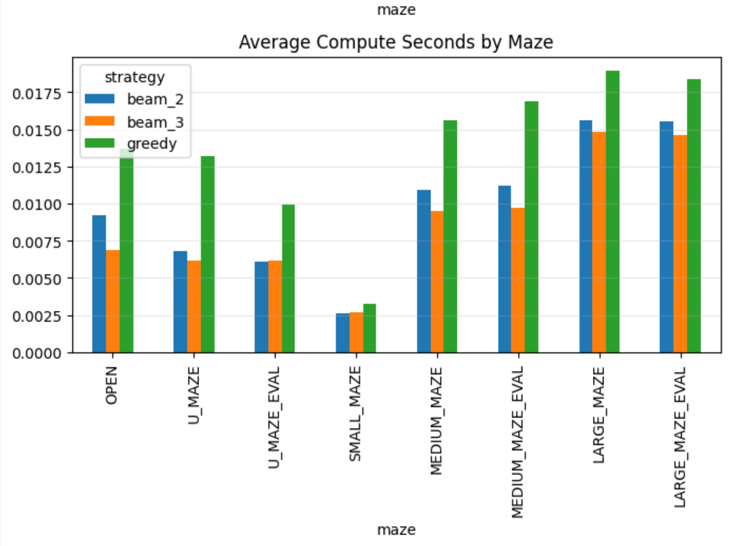
RESULTS



Strategy	Completion rate	Avg. steps to goal	Avg. compute time (s)	Avg. time per step (s)
beam_2	0.909973	7.077050	0.014012	0.001670
beam_3	0.975402	6.905816	0.012976	0.001735
greedy	0.691600	7.720542	0.017658	0.001547







FUTURE WORK

- Test how strongly the results depend on training on optimal trajectories.
- Compare additional decoding strategies beyond greedy and beam search.
- Explore combining model probabilities with reward-based guidance during decoding.

Q&A